

Machine Learning for IOH Prediction: VitalDB Dataset and Clinical AI

VitalWatch System Architecture & Clinical Decision Support Framework

1. The VitalDB Dataset

VitalDB is a publicly available high-resolution intraoperative vital signs database collected at Seoul National University Hospital. It contains records from 6,388 non-cardiac surgical patients and is one of the largest openly available surgical physiological datasets.

KEY DATASET CHARACTERISTICS:

- Sampling rate: vital signs recorded at 1/500 Hz to 500 Hz depending on device
- Parameters: arterial blood pressure (systolic, diastolic, MAP), heart rate, SpO2, end-tidal CO2, respiratory rate, temperature, neuromuscular blockade
- Cases: 6,388 adult surgical patients; 96 procedure types
- IOH label: binary label for MAP < 65 mmHg at each 1-minute window
- Patient split: patient-level train/test split prevents temporal leakage

The VitalWatch model uses a 29-feature engineered representation derived from raw MAP, HR, and SpO2 signals, including 1-minute and 3-minute rolling statistics, trend slopes, and clinically motivated binary indicators (e.g., map_below_75, map_dropping_fast, hr_map_ratio).

2. Logistic Regression Model Architecture

MODEL: Logistic Regression with L2 regularization (sklearn 1.6.1)

FEATURES (29 total):

Rolling statistics: MAP_mean_1m, MAP_mean_3m, MAP_std_1m, MAP_min_1m, MAP_min_3m

Trend features: MAP_trend_30s, MAP_trend_60s (linear regression slope over window)

HR statistics: HR_mean_1m, HR_mean_3m, HR_std_1m, HR_min_1m, HR_min_3m,

HR_trend_30s, HR_trend_60s

SpO2 statistics: SpO2_mean_1m, SpO2_mean_3m, SpO2_std_1m, SpO2_min_1m,

SpO2_min_3m, SpO2_trend_30s, SpO2_trend_60s

Clinical indicators: map_distance_to_65, map_below_75, map_below_70,
map_dropping_fast, hr_map_ratio

Current values: MAP_current, HR_current, SpO2_current

PERFORMANCE ON HELD-OUT PATIENT SET:

AUC-ROC: 0.91 (excellent discriminative ability)

At threshold 0.5: Precision ~0.37, Recall ~0.89, F1 ~0.52

Clinical interpretation: The model is tuned for high recall (sensitivity) — it catches ~89% of true IOH events at the cost of some false positives, which is clinically appropriate: missing an IOH event is worse than a false alarm.

3. SHAP Explainability and Clinical AI Principles

SHAP (SHapley Additive exPlanations) provides patient-level, instance-specific explanations for every prediction made by VitalWatch.

HOW SHAP WORKS FOR LOGISTIC REGRESSION:

- For linear models, SHAP uses a LinearExplainer with an independent masker
- Each feature receives a SHAP value = its additive contribution to the log-odds of the prediction, relative to the average prediction baseline
- SHAP values are signed: positive values push toward IOH prediction, negative values push away

CLINICAL INTERPRETATION:

- A SHAP value of +0.3 for MAP_trend_60s means: the patient's MAP decline over the past 60 seconds is adding 0.3 to the log-odds of IOH, increasing risk
- Features with the largest absolute SHAP values are the primary drivers for that specific patient at that specific moment in surgery
- This transforms the model from a "black box" to an auditable clinical tool

CLINICAL AI GOVERNANCE:

- VitalWatch provides decision support, not autonomous diagnosis

- Outputs are advisory; the attending anesthesiologist retains clinical authority
- All predictions are logged with input vitals, output risk score, and timestamp for retrospective audit
- The SHAP Risk Context panel must be visible at all times during monitoring

4. Interpreting Risk Scores and Alert Levels

RISK THRESHOLDS (configurable):

- STABLE (< 40%): Normal physiological variation; continue monitoring
- MEDIUM (40–60%): Elevated risk; heighten vigilance; consider pre-emptive intervention
- HIGH (> 60%): Imminent IOH predicted; recommend immediate intervention

RECOMMENDED CLINICAL RESPONSE BY LEVEL:

STABLE: No action required; document vital signs every 5 minutes

MEDIUM: Verify fluid status; check anesthetic depth; ensure vasopressor is drawn up and ready; increase MAP monitoring frequency to every 1–2 minutes

HIGH: Administer vasopressor (phenylephrine 100 mcg IV bolus first-line); reduce volatile anesthetic depth; accelerate IV fluid if hypovolaemic; call for attending presence if resident managing; document intervention

THE "5-MINUTE RULE": The model predicts risk in the upcoming 1-minute window.

Clinical response time from recognition to drug administration is typically 60–90 seconds.

A HIGH alert therefore provides approximately 3–4 minutes of actionable lead time before MAP would be expected to fall below 65 mmHg if untreated.